Postgraduate Program in

MULTI CLOUD ARCHITECTURE & DEVOPS

in collaboration with

Microsoft

Learn to automate application development life cycle on secure cloud infrastructure.



Hero VIRED | Virtual Education

aws certified Solutions Architect ASSOCIATE

AI GENAI for Devops

Career Assistance

POSTGRADUATE PROGRAM IN
MULTI CLOUD ARCHITECTURE & DEVOPS

DevOps is the combination of cultural philosophies, practices, and tools that increases an organization's ability to deliver applications and services at high velocity. This speed enables organizations to better serve their customers and compete more effectively in the market.

Cloud Engineering is the process of designing the systems necessary to leverage the power and economics of cloud resources to solve business problems.

The certificate program by Hero Vired in DevOps and Cloud Engineering, will help you build fundamental knowledge and skills to design, configure and deploy secure and robust cloud storage infrastructure.

The program explores the techniques of automating tasks and needs throughout the software development cycle, right from coding to testing and finally deployment. It integrates principles and practical hands-on experience needed to become a successful Cloud and DevOps Engineer.

in collaboration with

Microsoft

Hero VIRED | Virtual Education

POSTGRADUATE PROGRAM IN

MULTI CLOUD ARCHITECTURE & DEVOPS

The program starts with exploring advanced coding concepts in Python. You will learn collaborative coding techniques using version control systems and tools. Next, the program will introduce you to the DevOps environment that automates the CI/CD pipelines for rapid code build, testing, and deployment. The module includes the techniques and process involved in designing, configuring, and deploying cloud infrastructure and microservices for secure and economical application operability. Finally, the focus will shift to automating the monitoring and tracking process of the entire system.



Build skills in relevant DevOps and Cloud Engineering tools that are industry-acclaimed

circleci
GitHub
python
sonarqube
NGINX
Jenkins
docker
kubernetes
Terraform
ANSIBLE
Prometheus
kibana
Grafana
elasticsearch
aquasec trivy
Let's Encrypt
kafka
argo
redis
HELM

Cloud

Azure
Google Cloud Platform
aws
AI GENAI for Devops

DevSecOps, MLOps and DevOps Architecture, System Design & Cloud Architecture

PROGRAM HIGHLIGHTS



12+ INDUSTRY PROJECTS*

scenario deployments,
and case studies

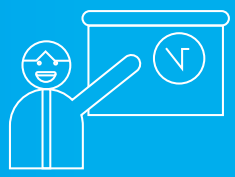


8 MONTH PROGRAM



BACHELOR'S DEGREE

required | **0-15 Years of Work Experience, Prior coding experience is mandatory**



60+

live sessions
with faculty from both
industry and academia
for holistic learning



Weekly effort of

12-15 HOURS

3 CLOUD PLATFORMS

Exposure to all the major cloud platforms **

aws Azure

*These projects are run internally by the mentors/faculty

** Terms and conditions apply

ELIGIBILITY CRITERIA

- Bachelor's degree is required
- Prior coding knowledge is mandatory
- Recommended work experience of 0-15 years

POTENTIAL JOB ROLES

TG	Possible Roles	Expected Salary	Aspirational Companies
0-5 Years	• Cloud Support Engineer • Cloud Engineer / DevOps Engineer	3-9 LPA	Google, amazon, Microsoft, TCS, wipro, HCL, accenture, SAP, PhonePe, IBM, Flipkart
6-10 Years	• Cloud Consultant • Site Reliability Engineer (SRE) • Platform Engineer • DevSecOps Engineer	8-28 LPA	
11-15 Years	• Cloud Solutions Architect • DevOps Architect • Cloud Security Architect	26-45 LPA	
15+ Years	• Principal SRE • Chief Cloud Architect • VP/Head of Cloud Engineering • Enterprise Cloud Strategist	50+ LPA	

Note - The career path for individuals post-completion of this program and with 3-5 years of work experience in DevOps would lead them to the following roles: DevOps Lead, Cloud Lead Engineer, Lead Platform Engineer, and Release Engineer.

N·S·D·C RE IMAGINE FUTURE

This program is accredited by the National Skill Development Corporation

CURRICULUM*

Duration: 8 months
Approximately 12-15 hours of student effort expected per week
Live projects integrated throughout the program duration

MODULE	TOPICS	SUB TOPICS
SDLC **(Self-paced)**	SDLC	• SDLC • The Waterfall Model • An Overview of the Agile Methodology • Transforming SDLC using Cloud and DevOps
Python Programming	Introduction to Python	• Loops • Conditional Statements
	Arrays, Sets, OOPs, etc.	• Oops • Classes • Sets • List
	Flask	• Flask • Routing
	Introduction to Database and Flask	• Database • MongoDB and PyMongo
Git and GitHub	Introduction to Git and Github	• Version Control using Git • Using GitHub for Collaborative Software Development
	Advance Git	• Git Merge • Branching • Git Reset
Agile **(Self-paced)**	Agile	• Agile Principles: Understand Effective Project Management • Scrum and Kanban: Apply Frameworks in Practice • Adaptive Leadership: Develop Skills in Agile Environments • Iterative Development: Implement Continuous Improvement Techniques • Team Collaboration: Foster Effective Communication • Problem-solving: Enhance decision-making abilities in Agile Projects

Note: AWS access will be provided for the program duration
*Curriculum is subject to changes

CURRICULUM*

MODULE	TOPICS	SUB TOPICS
Testing, Linux and Servers	Testing using Pytest (self-paced)	• Pytest Fundamentals • Test Case Creation • Test Automation • Result Analysis • Testing Efficiency Improvement
	Introduction to Linux and Commands	• Linux OS • Ubuntu
	File System and Permissions	• Ubuntu Commands • File System • Permissions
	Bash Programming	• Bash Scripts • Loops • Functions • Variables
	Understanding Networking	• OSI Model • TCP/IP • Routing
	Apache2, and Nginx	• Web Servers • Apache 2 • Nginx
	Project	• Understand the CI-CD Pipeline Workflow using traditional methodology (bash, python, git, etc.).
Cloud Essentials **(Self-paced)**	Introduction to Cloud Essentials	• SAAS • PAAS • IAAS

*Curriculum is subject to changes

CURRICULUM*

MODULES	TOPICS	SUB TOPICS
AWS Cloud Deployment	AWS Major Services	• Introduction to AWS IAM • EC2 • Elastic IPs • VPC • SSL Implementation
	AWS S3 and RDS	• RDS • S3 • Backups • Glacier
	AWS VPC	• 1. Understanding the VPC design • 2. Working with public-private subnet, NAT, IGW • 3. VPC Peering
	Elastic Load Balancer, AMIs and ASG	• Configuring and Managing AWS Elastic Load Balancer • Amazon Machine Images • Auto Scaling Groups (ASG) • Integrating ELB, AMIs, and ASG
	Project: Deploying MERN/ Python based Application in AWS	• AWS Account Setup • EC2 Instances • Networking Setup • Load Balancing • Auto Scaling • Security and Identity • Database Configuration • Storage and CDN • Deployment Automation • Monitoring and Logging
	AWS CloudFront and SSL	• AWS CloudFront • Creating and Configuring CloudFront Distributions • Content Delivery and Caching Strategies • Security and Access Control in CloudFront • Monitoring and Optimizing CloudFront Performance
Lightsail **(Self-paced)**	Lightsail	• Instance Management • Networking Configuration • Application Deployment

CURRICULUM*

MODULES	TOPICS	SUB TOPICS
Serverless Architecture & Cloud Automation	Automating using Boto3	• Boto3 Introduction • AWS Service Interactions • Scripting Automation
	AWS Lambda and Cli	• AWS CLI • Lambda • EBS
	AWS with GPT	• Develop practical skills in designing and implementing serverless solutions and automating using Generative AI
	AWS SNS and SQS	• AWS SES • SNS • SQS
AWS Elastic Beanstalk **(Self-paced)**	AWS Elastic Beanstalk	• Deploying python based application on Elastic Beanstalk
CI/CD Pipeline	Github Actions	• GitHub Actions • Software Development Flow
	Introduction to CI/CD Pipeline	• CI/CD Pipeline • Fundamentals of Pipeline
	Jenkins	• Jenkins • Groovy Code • Optimising Jenkins Pipeline • Configuring Agents • Caching • Pipeline optimization using Gen AI
AWS CodePipeline **(Self-paced)**	AWS CodePipeline	• AWS Codebuild • Codecommit • End-to-end Pipeline

CURRICULUM*

MODULES	TOPICS	SUB TOPICS
Containerization and Container Orchestration	Introduction to Containers and Docker	• Containerization • Docker
	Docker Hub and Docker Compose with GenAI	• Docker Hub • Docker Compose • Optimizing and automating Dockerfile using Gen AI
	ECR, ECS and Fargate	• AWS ECR • ECS • AWS Fargate
	Docker Swarm and Introduction to Kubernetes	• Docker Swarm, K8s
	Kubernetes	• Container Orchestration, K8s • Kubernetes Horizontal Pod Autoscaler (HPA), Vertical Pod Autoscaler (VPA • Kubernetes Services, Ingress Controllers (NGINX, Traefik) • GitOps, ArgoCD • Helm • EKS
	Deploying and Scaling Web Application	• Project
Automation	Terraform with GenAI	• IAC, and Terraform • Terraform states and environment management, Terraform infrastructure using GenAI
	Ansible	• Configuration Management using Ansible
	GitOps - ArgoCD	• GitOps • ArgoCD
	Project	• Terraform • Ansible
Cloudformation **(Self-paced)**	Cloudformation	• AWS CloudFormation • CloudFormation Stacks • CloudFormation Templates

CURRICULUM*

MODULES	TOPICS	SUB TOPICS
Monitoring	Graffana and Prometheus	• Graffana and Prometheus
	AWS Cloudwatch	• AWS Cloudwatch
	ELK Stack	• Elasticsearch • Logstash • Kibana

ARCHITECTURE

MODULES	TOPICS	SUB TOPICS
Architecture	AWS Architecture	• Understand key services like EC2, S3, RDS, VPC, and IAM and how they fit into AWS architecture. • Explore multi-AZ, load balancers, and auto-scaling concepts for resilient architecture. • Design a web-tier, app-tier, and database-tier using AWS services in a secure VPC setup.
	API & Deployment Strategies	• Create RESTful APIs using Amazon API Gateway and integrate with AWS Lambda for serverless execution. • Deployment Models: Blue/Green, Canary, Rolling • Redis & Kafka
	Optimizations & Recovery	• Use tools like AWS Cost Explorer, Trusted Advisor, and choose the right instance types and pricing models. • Leverage caching (CloudFront, ElastiCache) and compute scaling (ASG, Lambda) for better performance. • Implement snapshots, AMIs, cross-region replication, and recovery strategies using services like AWS Backup.
	Designing Architecture	• Use microservices, auto-scaling, and stateless design patterns for scalable architecture. • Security Best Practices: Design secure applications with IAM roles, security groups, KMS, and VPC endpoints. • Reference Architectures & Use Cases: Review real-world AWS architecture patterns (e.g., serverless, hybrid, and event-driven designs).
	Migration	• Migration Strategies • AWS Migration Tools : Explore tools like AWS Migration Hub, Application Discovery Service, and Server Migration Service (SMS). • Planning & Execution Steps: Cover pre-migration assessment, cutover planning, testing, and validation in a phased migration approach.

MICROSOFT AZURE

MODULES	TOPICS	SUB TOPICS
Azure	Introduction to Azure Administrator and Networking	• Overview of Azure core services, subscription structure, and the Azure Portal; manage users, roles, and access using Azure Active Directory (AAD). • Introduction to Azure virtual networking: create VNets, subnets, NSGs, and understand basic network routing and peering.
	Azure Databases & Storage	• Explore Azure Storage types: Blob, File, Queue, and Table storage, with lifecycle and access management. • Overview of Azure SQL Database, Cosmos DB, and basic provisioning, scaling, and security configurations.
	Azure DevOps	• Understand Azure DevOps services: Repos, Pipelines, Boards, and Artifacts for CI/CD lifecycle management. • Create a basic pipeline to build and deploy code using Azure Pipelines with GitHub or Azure Repos integration.
	Azure DevOps - 2	• Implement release pipelines, environment approvals, and deployment strategies (rolling, blue/green). • Integrate Infrastructure as Code (IaC) using ARM templates or Terraform with Azure Pipelines.
	Azure Monitoring and Governance	• Use Azure Monitor, and Log Analytics to track performance and troubleshoot resources. • Apply governance using Azure Policy and Cost Management to enforce compliance and budget controls.

GOOGLE CLOUD PLATFORM

MODULES	TOPICS	SUB TOPICS
GCP	Introduction to GCP and Networking	• Overview of GCP services, projects, billing, and resource hierarchy (organization, folders, projects). • Set up Virtual Private Cloud (VPC), subnets, firewalls, and understand GCP's global network architecture.
	GCP Storage and Databases and IAM	• Explore Cloud Storage, Cloud SQL for storing structured and unstructured data. • Manage user access and security using Identity and Access Management (IAM) roles and policies.
	Pipeline Architecture in GCP	• Design and build CI/CD pipelines using Cloud Build, Cloud Source Repositories, and Artifact Registry. • Automate deployments with Cloud Deploy or integrate GCP with GitHub Actions or Jenkins.
	GCP Monitoring and Logging	• Monitor infrastructure and applications using Cloud Monitoring, dashboards, and uptime checks. • Use Cloud Logging and Error Reporting to collect, filter, and analyze logs across GCP services.
	Serverless in GCP	• Build and deploy applications using serverless services like Cloud Functions, Cloud Run, and App Engine. • Understand event-driven architecture and integrate with Pub/Sub, Cloud Storage
CAPSTONE		

*Curriculum is subject to changes

LEARNER JOURNEY

Python boot camp - Build a simple api

Git and Github Coding Repo

Understanding Agile

Implement JIRA, GitHUb to simulate aAgile Project Code Management

Project

Automated Testing through PyTest

Linux shell scripting in Local machine using Linux VM

Deploy a Python Flask application in Linux Machine

Project

Deploy a Python Flask application in EC2 and Lambda

Project

AWS EC2, RDS and Lambda

Build a CICD pipeline withAutomated Testscripts

Project

Build a CICD Pipelining through Jenkins

Container Orchestration with Kubernetes

Deploy services using Docker in Kubernetes

Project

Work with Docker

Deploy application using IAC in Terraform

Project

IAC with Terraform and Ansible

Monitoring with Prometheus, Grafana

Project

Monitor the deployed application

Architecture & GenAI

AZURE & GCP Suite

Project

Implement the same tools in any other platform

SAMPLE PROJECTS



Architecting a 3-Tier Application on AWS VPC with ASG, ELB, and Backup Plan:

The objective of this project is to design and deploy a scalable and highly available 3-tier application on Amazon Web Services (AWS) Virtual Private Cloud (VPC). The architecture will include an Application Load Balancer (ALB) to distribute traffic, Auto Scaling Groups (ASGs) for automatic scaling, and a robust backup plan to ensure data resilience.



A Gaming Company:

Deploying and creating the scalable architecture of a live game which requires low latency between users coming from different parts of the world. The game server needs continuous updation initially (one update per week), so need to create CI-CD pipeline to update while having 0 downtime (Blue-Green deployment).



An EdTech Company:

Deploying a traditional LMS system created on MEAN stack. The architecture needs to be cost effective given many modules need higher computing resources occasionally while needing to control the cost of the usage of these resources.



Dockerized Microservices on Kubernetes: CI/CD with Jenkins and GitHub:

The Dockerized Microservices on Kubernetes project aims to build a microservices-based architecture using Docker containers and deploy them on Kubernetes. Additionally, the project focuses on automating the CI/CD pipeline using Jenkins and GitHub, providing a seamless development and deployment workflow. Microservices architecture promotes the decomposition of an application into smaller, loosely coupled services, enabling flexibility, scalability, and easier maintenance. Docker containers provide a lightweight and portable environment to package and isolate each microservice. Kubernetes, as a container orchestration platform, ensures efficient management and scaling of these microservices.



An OTT Platform:

Deploying OTT-based platform and scaling up to handle millions of concurrent users. The project includes the deployment architecture design to automate scaling up of microservice based applications while keeping the cost at check.

SAMPLE PROJECTS



Deploying Spring Boot Application on Kubernetes - AWS Project (End-to-End Web Application Deployment):

The Deploying Spring Boot Application on Kubernetes - AWS Project aims to deploy a Spring Boot web application on Kubernetes using AWS services, providing an end-to-end web application deployment solution. This project focuses on leveraging the power of Kubernetes for container orchestration and AWS services for seamless deployment.



Application Deployment and Scaling with Terraform AWS Three-Tier Architecture:

The Application Deployment and Scaling with Terraform AWS Three-Tier Architecture project automates the deployment and scaling of a three-tier application using Terraform and AWS. It leverages AWS services such as CloudFront/ELB, EC2/Fargate, RDS/DynamoDB, Auto Scaling, and CloudWatch for monitoring and logging. The project ensures security, follows best practices, and serves as a practical example for streamlining complex application deployments with Infrastructure as Code.

WORKSHOP

MODULES	TOPICS	SUB TOPICS
DevSecOps	Security in CI/CD Pipelines	• Integrate security tools (SAST, DAST, secrets scanning) into CI/CD pipelines using tools like SonarQube, Trivy, and GitHub Actions. • Automate security testing and vulnerability scans during build, test, and deployment stages. • Apply shift-left security practices to detect issues early in development.
	Governance, Compliance, and Runtime Protection	• Implement security policies to enforce compliance across infrastructure. • Use container runtime protection tools like Trivy and Falco for real-time threat detection. • Monitor and audit pipelines, containers, and cloud resources for anomalies and violations.
MLOps	Building ML Pipelines with DevOps Practices	• Build end-to-end ML pipelines using tools like MLflow, Kubeflow, or Vertex AI Pipelines. • Automate data preprocessing, training, evaluation, and versioning with Git and CI/CD tools. • Manage experiment tracking, artifact storage, and reproducibilit
	Deployment, Monitoring, and Governance	• Deploy ML models to production using Docker, Kubernetes, or cloud services like SageMaker or GCP AI Platform. • Monitor model performance with drift detection and logging (e.g., Prometheus, Grafana, or Evidently). • Implement model governance with version control, audit trails, and approval workflows.
GenAI with DevOps	Designing Gen AI Workflows	• Design GenAI application workflows integrating LLMs (OpenAI, Hugging Face) with APIs and data pipelines. • Use prompt engineering, chaining, and retrieval-augmented generation (RAG) to create effective workflows. • MCPs in DevOps
	Deploying and Scaling Gen AI with DevOps Practices	• Containerize and deploy GenAI applications using Docker, Kubernetes, or serverless platforms (Cloud Run, Lambda). • Automate deployment pipelines and scaling strategies for GenAI backends. • Monitor latency, usage, and token costs; apply rate-limiting and access control in production environments.

KEY FACULTY PROFILES



Dr. Prashant Kumar Dey

- Ph.D in Cybersecurity from CCU
- Master's Degree, Digital Security
- B.Tech, Electronics and Communications Engineering

Experience

Co-founder
AtharvaAI

Chief Technology Officer
Infosec United

Ex-Google, BMW



Rajan Chettri

- Masters in Computer Application | Sikkim Manipal Institute of Technology - SMU

Experience

Sr. SME (SRE / DevOps)
mthree

SME (Data Engineering)
Analytics Vidhya



Soumita Mukherjee

- MBA, Marketing from GIM, Goa
- Bachelor in Design from NIFT

Experience

Account Manager
Amazon

Marketing Manager
Pidilite Industries

Category Manager
HUL



LEARNING HOURS

Total Duration - 08 Months

150+ hrs	40+ hrs	~3
Total Live Sessions	Self Paced	Skill Test

Total weekly effort 12-15 hrs

YOUR LEARNER SUCCESS STORY



INDUSTRY FACULTY

Learners will have the opportunity to interact with domain experts from across industries through regular faculty lectures, industry sessions, and networking opportunities.



CAPSTONE PROJECT

All learners will work on multiple domain-focused real-world projects that will be spread out over the span of the program with expert mentoring and guidance sessions.



INDUSTRY PROJECT

Every learner will have the opportunity to gain industry experience through specialization-focused projects with the assistance of a mentor and an industry guide.



POST PROGRAM OUTCOMES

On completion of this program, you will be eligible for the following roles based on your work experience and skills: DevOps Engineer, Senior DevOps Engineer and specialist roles and skills like Cloud Engineer, Cloud Architect, Platform Engineer and Release Engineer.



CAREER SERVICES*

Each learner will work with assigned career specialist from the learner success team to build their personal brand, gain skills on stakeholder management, networking, problem-solving and build an entrepreneurial mindset.

*T&C apply as per the Hero Vired Career Services Policy

CERTIFICATION

On successful completion of the program,
you will be eligible for the following certificate



On successful completion of the program and clearing Microsoft Azure Exam,
you will be eligible for the following certificate



*Certificates are indicative and are subject to change. Terms and Conditions Apply.

CERTIFICATION



Additional Microsoft Azure Certification

(Subject to the learner clearing the certification examination)



*Certificates are indicative and are subject to change. Terms and Conditions Apply.

Note: Hero Vired will cover the costs of the Microsoft Azure Azure Fundamentals and AWS Certified Solutions Architect - Associate exams.

Gear Up for Greater Milestones

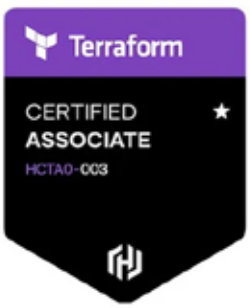
Industry Certification

This course will equip you with the knowledge and skills needed to excel in industry-recognized certification exams, ensuring you are ready to take on real-world challenges with confidence.

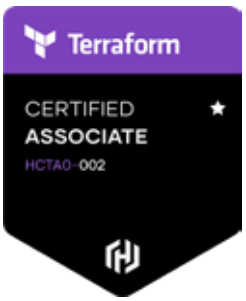












Please note: Hero Vired prepares students to achieve these certifications. However, the certification exam fees are not included in the course fee and must be paid separately by the candidates if they choose to take the exam.

WHY VIRED?

Hero Vired offers a premium learning experience with industry relevant programs and world class partnerships, to create the change makers of tomorrow.

With a goal to make learning a continuous endeavour, Vired offers a holistic experience that is applied, real-time and engaging.

At Hero Vired, we believe everyone is made for big things. With the experience, knowledge and expertise of the Hero Group, Hero Vired is on a mission to change the way we learn.



1

VIL – Vired Integrated Learning

Comprehensive professional development through measurable transformation

2

VIR - Vired Industry Ready

'Industry relevant focus' from the industry, of the industry, for the industry

3

I Vired – Vired Personalized Training

Personalized & high-engagement learning experience

4

Vired Access

Premium international learning experiences

5

Vired Globally

Recognized certification from the world's renowned institutions

OUR PARTNERS













































































THE HERO GROUP IN EDUCATION

The Hero Group has made significant contributions in the field of K12, medical education and higher education.

IN PRIMARY AND HIGHER SECONDARY EDUCATION

Raman Munjal Vidya Mandir

BCM Chain of Schools

Green Meadows School

IN HIGHER EDUCATION

ISB Founding Members

BML Munjal University

Dayanand Medical College & Hospital

THE HERO STORY

The Hero Group is one of the **leading business conglomerates in the world.** The company saw its humble beginning in 1956 when the four Munjal brothers migrated to Ludhiana from Kamalia (now in Pakistan). As first-generation entrepreneurs, they started out by manufacturing bicycle components and then began rapidly expanding the business. From there, they continued their growth story by diversifying and deepening their expertise across domains.

Today, the **US $5 billion diversified Hero Group** is a conglomerate of Indian companies **with primary interests and operations in automotive manufacturing, financing, renewable energy, electronics and education.**

Hero VIRED | Virtual Education

Wait Nahi, *Great* Kar!

Want more information on the program?

Reach us at **1800 309 3939** | Visit us at **www.herovired.com**